Assessment 3

Feature Implementation & Analysis

Charts Integration

Sound Pollution Hunter

Multi-language

Enhancing Educational Impact through Technology

Core Goal

The Sound Pollution Hunter app is an interactive platform where students collect, visualize, and analyze environmental sound data.

Primary Features

Dynamic charts for pattern identification and multi-language support for global accessibility.

Transformational Impact: These features evolve the app from a simple data entry tool into a comprehensive educational platform.

"Visualizing data is the another step toward scientific understanding."

Technical Foundation: Installation & Setup

NPM Ecosystem

Extending base functionality with industry-standard libraries. Main ones

```
npm install recharts
npm install i18n-js
npm install expo-localization

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Clone the Github Project
cd [Github-project]

npm i
```

Key Libraries

Recharts: High-quality data visualization.

i18n-js: Robust translation management.

Expo Localization: Native locale detection.

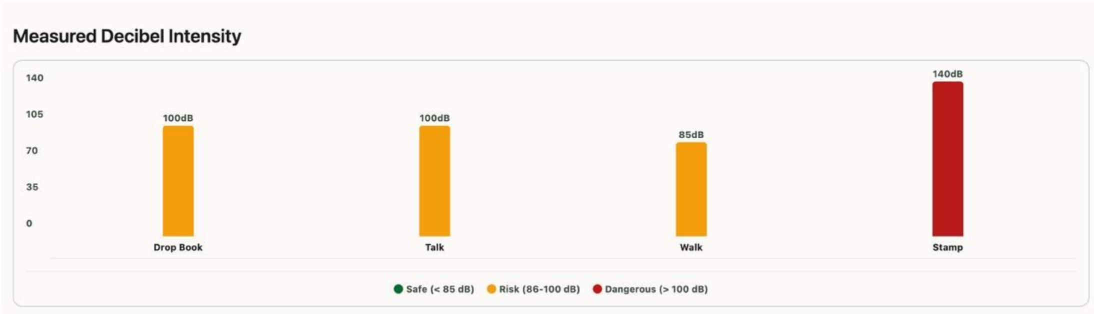
Charts implementation

Technical Integration

- Reusable components accepting dynamic JSON data.
- Real-time rendering in the **Journal Tab**.
- Optimized for mobile-responsive layouts.

Educational Value

- Measured decibel intensity
- Visual pattern identification for students.
- Development of scientific analysis skills.



Accessibility: Multi-language Support (i18n)

Technical Strategy

- Uses static JSON files for different locales.
- Managed through a central translation hook.
- Follows industry-standard i18n best practices.

Accessibility Impact

Directly addresses barriers for non-native speakers, making the app inclusive for a global student body.
Current implementation: English & Indonesian

Inclusive Design | Global Reach

Dynamic UI Updates

The translation system ensures a seamless user experience:

- Instant language switching in Settings.
- No app restart required for text updates.
- Zero latency using local resources.

"Accessibility is not a feature, it's a fundamental requirement."

Technical Feasibility Analysis

Implementation Strengths

→ **High Feasibility:** Reliance on established, well-documented libraries with strong community support.

→ **Efficient Translation:** System is lightweight, using local JSON resources for zero-latency UI updates.

→ **Modular Charts:** Reusable component structure allows for easy scaling across different experiment types.

Technical Challenges

→ **Platform Adaptation:** Recharts is web-centric; adapting it for React Native requires specific configuration.

→ **Rendering Complexity:** Ensuring consistent visual behavior across iOS and Android environments.

→ **Data Handling:** Managing manual inputs vs. automated library expectations for precise rendering.

"Feasibility is high, but requires careful cross-platform optimization."

Performance and Compatibility Insights

Minimal Impact

- **Translation Overhead:** Negligible impact as static resources are loaded locally.
- **Zero Latency:** Instant UI updates without network requests.

Scalability

- **Efficient Rendering:** Optimized for small-to-medium datasets typical of student experiments.
- **Future-Proofing:** Large data inputs would require sampling to maintain smooth UI performance.

Platform Parity

- **Expo Support:** Strong cross-platform consistency across iOS and Android.
- **Visual Adjustments:** Minor CSS/style tweaks needed for chart libraries between platforms.

"Consistent performance is key to a reliable educational experience."

Accessibility and Ethical Considerations

Inclusive Design

- **Linguistic Inclusivity:** Multi-language support removes barriers for non-native speakers and international students.
- **Visual Learning:** Dynamic charts provide an alternative data processing method for visual learners.
- **User Empowerment:** Features are designed to make complex data accessible to all student demographics.

Ethical Responsibility

- **Data Precision:** Since data is manually input, the app must be framed as a learning tool, not a scientific instrument.
- **Accuracy Awareness:** Users are encouraged to understand the limitations of manual measurement systems.

"Ethical technology use involves transparency about a tool's limitations and its intended educational purpose."

Strategic Value: Why These Features?

Usability

- **Visual Feedback:** Charts make complex data more digestible and engaging for students.
- **Intuitive Flow:** Seamless transition from data entry to visualization.

Accessibility

- **Inclusivity:** Multi-language support opens the app to a global student body.
- **Visual Learning:** Charts provide an alternative way to process information.

Educational Value

- **Critical Thinking:** Encourages students to compare predictions with actual data.
- **Scientific Analysis:** Helps in identifying environmental sound patterns.

"These features significantly improve usability, accessibility, and educational value."